

Deepa Kale

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SUMMARY

Computer Science graduate from Texas Tech University with hands-on experience in AI/ML, full-stack software development, and database systems. Built machine learning solutions using TensorFlow and Python, leveraging Cursor AI throughout development, and developed scalable web applications using React, Node.js, Express, and MongoDB. Experienced in Agile development, REST API design, and AI-assisted software engineering workflows.

EDUCATION

Texas Tech University, Lubbock, Texas <i>Bachelor of Science in Computer Science</i>	May 2026 GPA: 3.5
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ACADEMIC PROJECTS

Senior Capstone - TCP Traffic Anomaly Detection System <i>Python, TensorFlow, Pandas, Scikit-Learn</i>	2025
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- Designed and developed an AI-powered network traffic anomaly detection platform using Python, TensorFlow, and machine learning techniques to identify malicious TCP traffic patterns across 800,000+ network records
- Built an end-to-end data pipeline that cleaned, normalized, and transformed raw TCP log data, extracted TCP handshake features, and generated structured datasets for model training and inference
- Trained and optimized a CNN-based anomaly detection model over 100 epochs, achieving 94–96% training accuracy and 92–94% prediction accuracy, while evaluating performance using precision, recall, confusion matrices, and validation metrics
- Developed a modular command-line application that automated log ingestion, TCP handshake verification, AI inference, and anomaly reporting, exporting CSV summaries and visualization reports

Custom Apparel E-Commerce Platform <i>React, JavaScript, Tailwind CSS, Node.js, Express.js, MongoDB</i>	2025
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- Developed a full-stack e-commerce web application with product browsing, shopping cart functionality, inventory management, and backend API integration using React, Node.js, Express.js, and Tailwind CSS
- Designed RESTful API endpoints for product retrieval, cart operations, image handling, and frontend-backend communication using Express.js
- Structured backend logic using modular Express.js controllers to separate routing, business logic, and database interactions, improving maintainability and supporting future feature expansion
- Created MongoDB document schemas for products and transactions, designing flexible data models to support evolving product attributes while maintaining efficient querying and scalable data management

Hurricane Relief Resource Management Database (HEART) <i>SQL, MySQL, Database Design</i>	2025
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- Designed a relational database system to support emergency response operations by managing shelters, evacuees, volunteers, resources, and allocation records
- Created ER diagrams and normalized database schemas through Third Normal Form (3NF), defining primary keys, foreign keys, and constraints across entities to reduce redundancy and maintain data integrity
- Developed SQL queries using CTEs, JOINS, aggregations, filtering conditions, and subqueries to generate operational reports for shelter occupancy, volunteer assignments, resource distribution, and emergency response tracking

AI-ASSISTED DEVELOPMENT

- Leveraged GitHub Copilot and generative AI tools to accelerate software development, debugging, code refactoring, documentation generation, and test case creation
- Used Cursor to debug and resolve errors during full-stack development of the e-commerce platform, accelerating issue resolution and improving code quality
- Applied AI-assisted workflows to improve productivity in full-stack application development and machine learning experimentation

LEADERSHIP & EXPERIENCE

Student Development and Engagement <i>Peer Mentor, Raider Ready Program & Tech Leadership Institute.</i>	Lubbock, TX January 2023 – June 2024
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- Mentored 40+ students through structured academic and career support, helping students navigate university resources and achieve personal goals
- Developed Excel-based tracking systems to monitor student attendance, mentoring progress, program participation, and SurveyMonkey student satisfaction data across Raider Ready and Tech Leadership Institute programs

TECHNICAL SKILLS & COURSEWORK

Developer Tools: Git, GitHub, VS Code, Xcode, JupyterLab, MATLAB

AI Tools: Cursor, Claude Code, GitHub Copilot

Programming Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL, HTML, CSS

Development Practices: Agile Methodologies, Scrum, Sprint Planning, Software Development Life Cycle (SDLC)

Frameworks & Technologies: React, Node.js, Express.js, Tailwind CSS, REST APIs

Databases: MySQL, PostgreSQL, MongoDB, Relational Database Design, Query Optimization

Coursework: Operating Systems, Database Systems, Computer Organization & Assembly Language, Data Structures & Algorithms, Applied Data Science, Machine Learning, Object-Oriented Programming